

Technical note: the zero crossing of the two-point correlation function in environment-split clustering

ASTRA clustering project

June 2026

Abstract

We measure the two-point correlation function (2PCF) of an AbacusSummit HOD galaxy catalog, split by cosmic-web environment with the ASTRA classification. All our statistics — the full-sample auto-correlation, the environment-quantile auto-correlations, and the cross-correlations with the full sample — change sign at the same scale, $s \simeq 120 h^{-1} \text{Mpc}$. This note explains why this zero crossing exists, why its location is shared by all linearly biased tracers, and how our finite subbox volume shifts it. The crossing scale is a known cosmological standard ruler related to the horizon size at matter–radiation equality [1].

1 Why the correlation function must cross zero

The correlation function is the Fourier transform of the power spectrum,

$$\xi(r) = \frac{1}{2\pi^2} \int_0^\infty dk k^2 P(k) j_0(kr), \quad (1)$$

where $j_0(x) = \sin x/x$ is the spherical Bessel function. The inverse relation at $k = 0$ gives a sum rule,

$$P(k=0) = \int \xi(r) d^3r. \quad (2)$$

For a primordial spectrum $P(k) \propto k^{n_s}$ with $n_s > 0$, the left-hand side of Eq. (2) vanishes. The strong positive correlations at small r must therefore be compensated by a shallow negative tail at large r . In other words, every overdense region sits inside a slightly underdense larger environment. The scale where $\xi(r)$ changes sign is set by the turnover of $P(k)$, which reflects the size of the horizon at matter–radiation equality. For Planck-like parameters the linear-theory crossing lies at $r_0 \approx 125\text{--}130 h^{-1} \text{Mpc}$, just outside the BAO peak. Reference [1] showed with N -body simulations that this crossing is stable against nonlinear evolution, redshift-space distortions, and galaxy bias, and proposed it as a standard ruler for the equality horizon. A related, template-free ruler built from features of $\xi(r)$ near the BAO range is the “linear point” [2, 3].

2 Why all biased tracers cross at the same scale

At leading order, a galaxy sample g with linear bias b follows the matter field,

$$\xi_g(r) = b^2 \xi_m(r), \quad \xi_{g_1 g_2}(r) = b_1 b_2 \xi_m(r). \quad (3)$$

In redshift space, linear theory (Kaiser) modifies only the amplitude of the monopole,

$$\xi_0(s) = (b^2 + \frac{2}{3}bf + \frac{1}{5}f^2) \xi_m(s), \quad (4)$$

with f the linear growth rate. Equations (3) and (4) rescale ξ but cannot move its zero. As a consequence, all tracers with scale-independent bias must cross zero at the same comoving scale, at any redshift. This makes the common crossing a simple consistency test: if an environment-selected sample crossed at a clearly different scale, this would signal scale-dependent bias.

3 Our setup

We use one AbacusSummit base box (cosmology c000, phase ph000, side $2000 h^{-1}$ Mpc) [4], populated with an LRG-like HOD at $z = 0.5$ (catalog `hod484` from the EMC emulator set, 4×10^6 galaxies). Positions include redshift-space distortions along the z axis. We divide the box into a $4 \times 4 \times 4$ grid of 64 subboxes of $500 h^{-1}$ Mpc and treat each subbox as an independent volume with open boundaries.

In each subbox, the ASTRA method classifies the local environment. We build a Delaunay triangulation over the union of the galaxies and an equal number of uniform random points. For every galaxy we count data neighbours n_d and random neighbours n_r in the triangulation and define the local density indicator

$$r_{\text{ASTRA}} = \frac{n_d - n_r}{n_d + n_r}. \quad (5)$$

Galaxies and randoms are then split independently into four quantiles of r_{ASTRA} (Q1 = most underdense, Q4 = most overdense). We measure the 2PCF multipoles ($\ell = 0, 2$) with `pycorr/Corrfunc` [6] in 15 separation bins over $0 < s < 150 h^{-1}$ Mpc, using the Landy–Szalay estimator [5]

$$\hat{\xi} = \frac{DD - 2DR + RR}{RR}, \quad (6)$$

with a fixed catalog of uniform geometry randoms ($5 \times$ the data). The mean over the 64 subboxes gives the signal; the scatter gives the cosmic variance.

4 Results

Figure 1 shows the full-sample auto-correlation. Figure 2 shows the auto-correlations of the four data and random quantiles, and Figure 3 the cross-correlations of each quantile with the full sample. Table 1 lists the measured zero crossings of the monopole. All statistics with a positive-to-negative crossing agree on $s_0 \simeq 117\text{--}123 h^{-1}$ Mpc. The Q1 cross-correlation is negative at small s (underdense regions are anti-biased) and approaches zero from below, so it has no such crossing.

The common crossing is consistent with the linear-bias expectation of Section 2. Its location is a few h^{-1} Mpc *inside* the linear-theory value. We attribute this to the integral constraint of the finite subbox: the estimator of Eq. (6) measures clustering with respect to the mean density of each $500 h^{-1}$ Mpc subbox, which biases the estimate low by roughly the volume average

$$\hat{\xi}(s) \approx \xi(s) - \bar{\xi}_V, \quad \bar{\xi}_V = \frac{1}{V^2} \int_V \int_V \xi(|\mathbf{x} - \mathbf{y}|) d^3x d^3y \sim 10^{-3}. \quad (7)$$

At $s \gtrsim 100 h^{-1}$ Mpc the true $|\xi|$ is of the same order as $\bar{\xi}_V$, so the constraint pulls the apparent crossing inward and deepens the negative tail. Averaging over subboxes does not remove this effect because it acts inside each subbox. A measurement in the full $2000 h^{-1}$ Mpc box would move the crossing back toward the linear-theory value.

Statistic	auto s_0 [h^{-1} Mpc]	cross with full s_0 [h^{-1} Mpc]
Full sample	123	—
Q1 (most underdense)	135	no $+\rightarrow-$ crossing
Q2	112	117
Q3	123	123
Q4 (most overdense)	123	123

Table 1: Zero crossings of the monopole, from the mean over 64 subboxes of the c000 (fiducial) run. The statistical precision per bin near the crossing is a few h^{-1} Mpc.

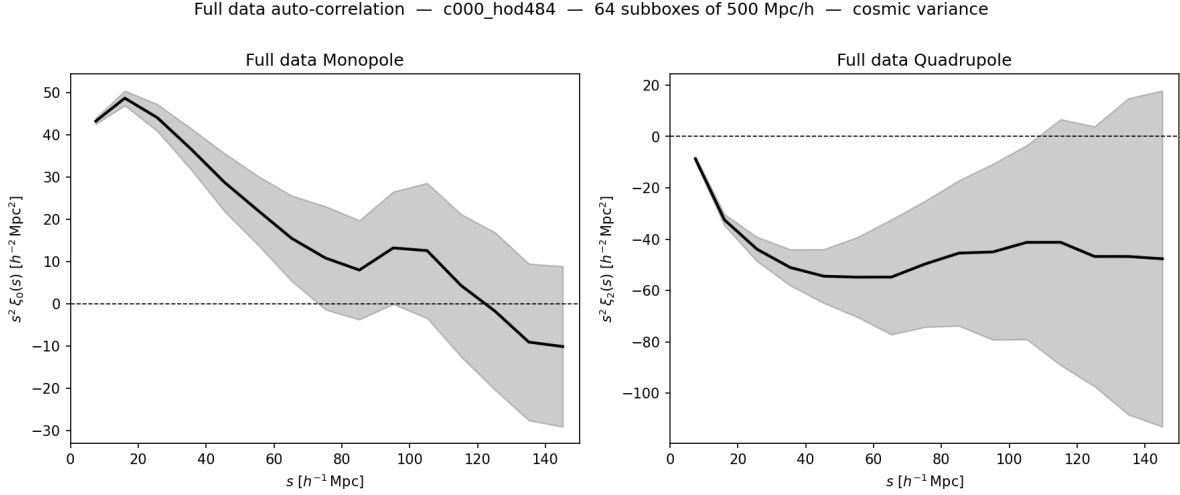


Figure 1: Full-sample auto-correlation (monopole and quadrupole), mean and $\pm 1\sigma$ over 64 subboxes. The monopole crosses zero at $s \simeq 123 h^{-1}$ Mpc.

5 Outlook

Two points follow for our Fisher-matrix program. First, the crossing scale tracks the equality horizon, so it shifts with ω_m ; part of the information in our ω_c derivatives will come from this feature. Second, because the crossing is invariant under scale-independent bias, any future displacement of the quantile crossings relative to the full sample would be a clean detection of scale-dependent environmental bias, not a systematic error.

References

- [1] F. Prada, A. Klypin, G. Yepes, S. E. Nuza, S. Gottlöber, *Measuring equality horizon with the zero-crossing of the galaxy correlation function*, arXiv:1111.2889 (2011).
- [2] S. Anselmi, G. D. Starkman, R. K. Sheth, *The linear point: a cleaner cosmological standard ruler*, arXiv:1703.01275 (2017).
- [3] S. Anselmi et al., *Galaxy Correlation Functions Provide a More Robust Cosmological Standard Ruler*, Phys. Rev. Lett. 121, 021302 (2018).

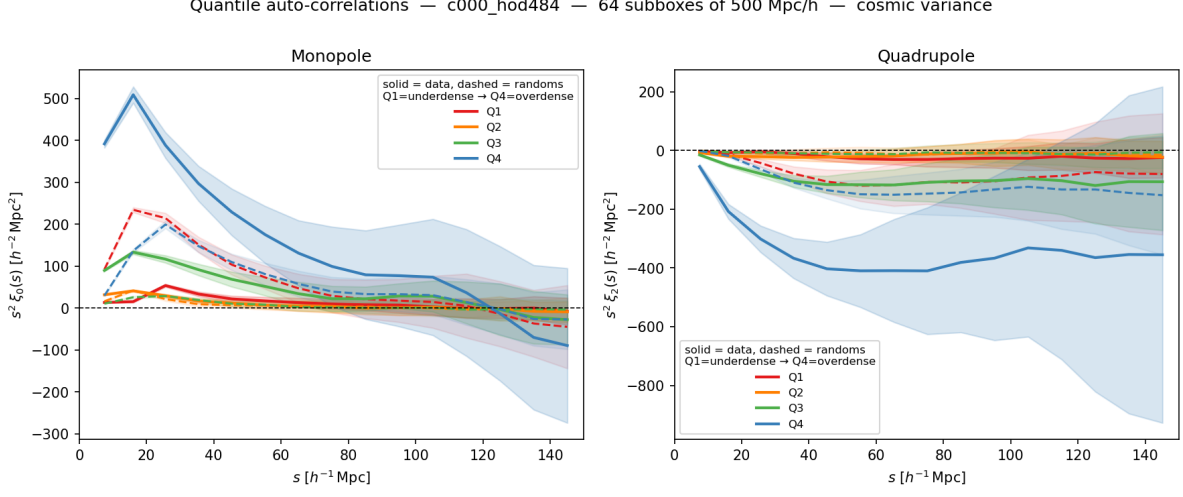


Figure 2: Auto-correlations of the four ASTRA quantiles for galaxies (solid) and ASTRA randoms (dashed). Despite very different amplitudes, the crossings agree within the errors.

- [4] N. A. Maksimova et al., *AbacusSummit: a massive set of high-accuracy, high-resolution N-body simulations*, MNRAS 508, 4017 (2021).
- [5] S. D. Landy, A. S. Szalay, *Bias and variance of angular correlation functions*, ApJ 412, 64 (1993).
- [6] M. Sinha, L. H. Garrison, *Corrfunc: a suite of blazing fast correlation functions on the CPU*, MNRAS 491, 3022 (2020).

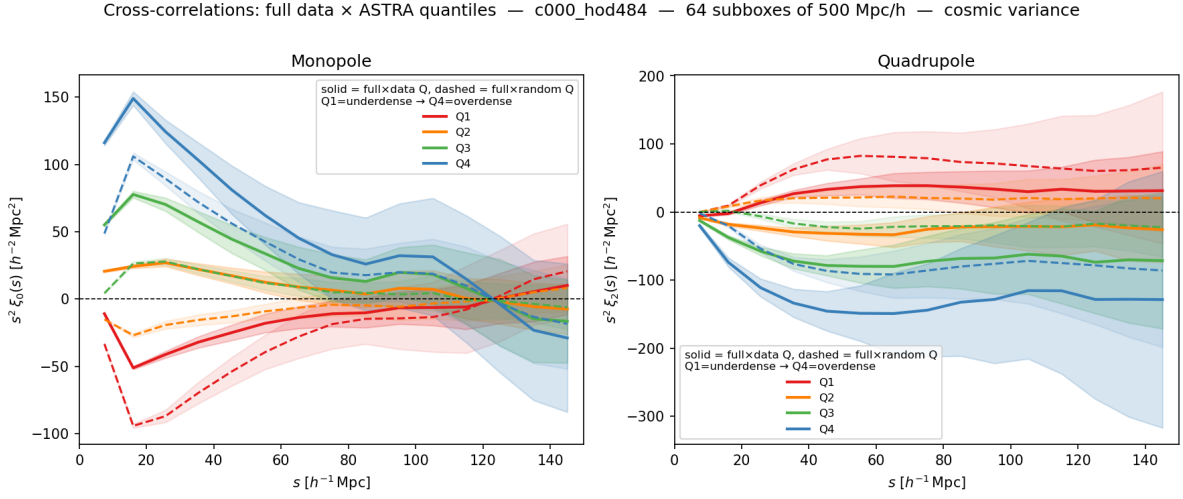


Figure 3: Cross-correlations of the full sample with each data quantile (solid) and each random quantile (dashed). Q2–Q4 cross zero at the same scale as the full sample; Q1 approaches zero from below.